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United States Patent
Kind Code
Date of Patent
Inventor(s)

12418902
B2
September 16, 2025
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Private network management including licensed and unlicensed radio unit channel aggregation orchestration

Abstract

A method of operating a private network includes obtaining channel requirement information indicating a bandwidth of a channel, obtaining available channel information indicating Citizens Broadband Radio Service (CBRS) frequency bands and non-CBRS frequency bands available to be allocated to the channel, selecting one or more of the CBRS frequency bands and one or more of first non-CBRS frequency bands based on the channel requirement information, generating channel aggregation information indicating the selected CBRS and non-CBRS frequency bands, generating transmission power level information for the selected CBRS and non-CBRS frequency bands, transmitting the channel aggregation information and the transmission power level information, receiving the first channel aggregation information and the first transmission power level information at an access point device, and transmitting data in the channel using the selected CBRS and non-CBRS frequency bands from the access point device based on the first transmission power level information.

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Appl. No.: 17/975325

Filed: October 27, 2022

Prior Publication Data

Document Identifier

Publication Date

Publication Classification

Int. Cl.: H04W72/0453 (20230101); H04W16/14 (20090101); H04W84/12 (20090101)

U.S. Cl.:

CPC H04W72/0453 (20130101); H04W16/14 (20130101); H04W84/12 (20130101)

Field of Classification Search

USPC: None

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Background/Summary

BACKGROUND

(1) Conventionally, cellular access point devices communicate with client devices using cellular communications standards (e.g., 4G LTE or 5G communications standards from the 3rd Generation Partnership Project (3GPP)). Also, wireless local area network (WLAN) access point devices communicate with client devices using WLAN communications standards (e.g., Wi-Fi using one or more of the 802.11 family of communications standards from the Institute of Electrical and Electronics Engineers (IEEE)). Conventionally, cellular access point devices are provided separately from WLAN access point devices and, thus, channel allocation for cellular communications is done separately from channel allocation for WLAN communications. In addition, management of cellular access point devices is done separately from management of WLAN access point devices.

BRIEF SUMMARY

(2) According to the present disclosure, private networks are formed from access point devices that communicate with client devices using both cellular communications standards and WLAN

communications standards. Also, a single network management platform manages both channels in which data is communicated using cellular communications standards and channels in which data is communicated using WLAN communications standards. In addition, a channel allocation device aggregates both licensed frequency bands and unlicensed frequency bands for channels based on requirements (e.g., Quality of Service (QoS) requirements) of client device and applications.

(3) A method of operating a private network according to a first aspect of the present disclosure includes: obtaining, by a channel allocation device, first channel requirement information indicating a first bandwidth of a channel; obtaining, by the channel allocation device, available channel information indicating a plurality of first Citizens Broadband Radio Service (CBRS) frequency bands and a plurality of first non-CBRS frequency bands available to be allocated to the channel; selecting, by the channel allocation device, one or more of the first CBRS frequency bands and one or more of the first non-CBRS frequency bands based on the first channel requirement information; generating, by the channel allocation device, first channel aggregation information indicating the one or more of the first CBRS frequency bands and the one or more of the first non-CBRS frequency bands selected based on the first channel requirement information; generating, by the channel allocation device, first transmission power level information indicating one or more first transmission power levels respectively corresponding to the one or more of the first CBRS frequency bands and one or more first transmission power levels respectively corresponding to the one or more of the first non-CBRS frequency bands; transmitting, by the channel allocation device, to an external device, the first channel aggregation information and the first transmission power level information; receiving, by a first access point device, the first channel aggregation information and the first transmission power level information; and transmitting, by the first access point device, first data in the channel using the one or more of the first CBRS frequency bands and the one or more of the first non-CBRS frequency bands based on the first transmission power level information.

(4) A method of operating a private network according to a second aspect of the present disclosure includes: obtaining, by a control device, access point information indicating one or more access point devices including a first access point device; obtaining, by the control device, client device information indicating a plurality of client devices including a first client device and a second client device; generating, by the control device, channel information indicating a plurality of channels, including a first channel and a second channel; transmitting, by the first access point device, to the first client device, first data according to a cellular communication standard using the first channel; and transmitting, by the first access point device, to the second client device, second data according to a Wireless Local Area Network (WLAN) communication standard using the second channel.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Non-limiting and non-exhaustive embodiments are described with reference to the following drawings. In the drawings, like reference numerals refer to like parts throughout the various figures unless otherwise specified.

(2) For a better understanding of the present disclosure, reference will be made to the following Detailed Description, which is to be read in association with the accompanying drawings.

(3) FIG. 1 is a diagram of a communication system in accordance with embodiments described herein.

(4) FIG. 2 is a block diagram illustrating an example of an access point device in accordance with embodiments described herein.

(5) FIG. 3 is a block diagram illustrating an example of a private network control device in accordance with embodiments described herein.

(6) FIG. 4 is a block diagram illustrating an example of a channel allocation device in accordance with embodiments described herein.

(7) FIGS. 5A and 5B illustrate a logical flow diagram showing an example of a method of operating a private network in accordance with embodiments described herein.

(8) FIGS. 6A and 6B illustrate a logical flow diagram showing an example of a method of operating a private network in accordance with embodiments described herein.

DETAILED DESCRIPTION

(9) According to the present disclosure, a novel private network small cell is provided by one or more access point devices that utilize a unique combination of cellular frequency bands and Wireless Local Area Network (WLAN) frequency bands (e.g., Wi-Fi frequency bands). For example, a combination or carrier aggregation of 4.sup.th Generation (4G) Long Term Evolution (LTE) cellular bands and 5.sup.th Generation (5G) New Radio (NR) cellular bands, such as mid-band spectrum bands, b48, n77, and n48 (i.e., Citizens Broadband Radio Service (CBRS) frequency bands), and unlicensed frequency bands (e.g., Wi-Fi frequency bands) enables greater bandwidth of licensed spectrum.

(10) Because of the difference in FCC rules of operation for CBRS frequency bands (i.e., n48 frequency bands) versus non-CBRS frequency bands (e.g., n77 frequency bands) and unlicensed frequency bands (e.g., Wi-Fi frequency bands such as 2.4 GHz, 5 GHz, and 6 GHz frequency bands), channel aggregation orchestration is required in order to combine CBRS frequency bands, non-CBRS frequency bands, and unlicensed frequency bands. Although 3rd Generation Partnership Project (3GPP) standards exist (e.g., 3GPP TS 38.101-1 V15.18.0 (2022-06)) that establish the various carrier aggregation combinations of the n77 and n48 frequency bands, the present application teaches an orchestration mechanism that selects and deploys specific combinations in real-time for a given network payload or QoS requirement, which is not addressed by existing 3GPP standards. Moreover, there is no known solution that orchestrates channels formed by aggregating n48 frequency bands and n77 frequency bands.

(11) While n77 frequency bands are mentioned as an example, orchestration of the use of frequency bands other than n77 frequency bands is within the scope of the present disclosure. For example, the orchestration mechanism disclosed herein could be used for other combinations of frequency bands such as n48 frequency bands and n71 frequency bands. Also, orchestration of G frequency bands (e.g., 110 GHz to 300 GHz) is within the scope of the present disclosure.

(12) If WLAN frequency bands (e.g., Wi-Fi frequency bands) are used, a transmission using those frequency bands is subject to a much lower power output limit than transmissions in both n48 frequency bands and n77 frequency bands, for example. Access point devices according to the present disclosure employ a Radio Unit (RU) design that leverages any combination of spectrum that includes CBRS frequency bands (i.e., n48 frequency bands) with other bands, including unlicensed frequency bands (e.g., Wi-Fi frequency bands), an RU channel aggregation orchestration mechanism manages the operation of the RU to adhere to Federal Communication Commission (FCC) operating rules while still meeting network performance requirements.

(13) By way of non-limiting example, channel aggregation orchestration is performed using a combination of unlicensed frequency spectrum and licensed frequency spectrum. Such unlicensed frequency spectrum includes WLAN frequency spectrum (e.g., Wi-Fi frequency bands, including 2.4 GHz, 5 GHz, and 6 GHz frequency bands), and the CBRS General Authorized Access (GAA) frequency spectrum (e.g., frequency bands in the 3.55-3.70 GHz range). Such licensed frequency spectrum includes CBRS frequency bands (e.g., n48 frequency bands in the 3.55-3.65 GHz range), Priority Access License (PAL) frequency bands, non-CBRS frequency bands (e.g., n77 frequency bands in the range of 3.45-3.55 GHz), etc.

(14) According to the present disclosure, WLAN protocols (e.g., Wi-Fi protocols) and cellular protocols (e.g., 5G/4G LTE protocols) are used by access point devices for communication with end user or client devices (e.g., User Equipment (UE) devices such as phones, tablets, adapters

connected to industrial machinery, Augmented Reality (AR)/Virtual Reality (VR) headsets, etc.). Access point devices according to the present disclosure can utilize cellular protocols (e.g., 5G/4G LTE protocols) and WLAN protocols (e.g., Wi-Fi protocols) for fronthaul communications with end user or client devices. In addition, small cell private networks according to the present disclosure can utilize either WLAN protocols (e.g., Wi-Fi small cell private networks) or cellular small cell private networks protocols (e.g., 5G/4G LTE small cell private networks) for wireless communication between access point devices that are linked together to form a small cell. The present disclosure teaches a single platform capable of integrated network management that enables such private networks to be created and managed. Accordingly, a single platform is provided that can manage all devices, configurations, and policies within a private network, unlike conventional solutions that require separate platforms and, thus, separate management of configurations and policies for cellular devices and WLAN devices.

(15) FIG. 1 is a diagram of a communication system **100** in accordance with embodiments described herein. The communication system **100** includes a private network **102** that communicates with a private network core **104** via a network **106**. In one or more implementations, the network **106** is provided by a commercial Internet Service Provider (ISP).

(16) The private network **102** is provided by access point devices **108-1**, **108-2**, and **108-3**. Although the private network **102** shown in FIG. 1 includes three access point devices, private networks that include a different number of access point devices are within the scope of the present application.

(17) Each of the access point devices **108-1**, **108-2**, and **108-3** communicates with at least one other of the access point devices **108-1**, **108-2**, and **108-3**. Also, each of the access point devices **108-1**, **108-2**, and **108-3** communicates with one or more client devices. The client devices may include a combination of different types of devices, such as cell phones or User Equipment (UE) devices, Augmented Reality (AR)/Virtual Reality (VR) headsets, security cameras, tablet computers, laptop computers, and wireless network adapters for industrial machinery, for example.

(18) In the example shown in FIG. 1, the access point device **108-1** communicates with the access point devices **108-2** and **108-3**. Also, the access point device **108-1** communicates with client devices **110-1**, **110-2**, and **110-3**. The access point device **108-2** communicates with client devices **110-4**, **110-5**, and **110-6**. The access point device **108-3** communicates with client devices **110-7**, **110-8**, and **110-9**.

(19) In one or more implementations, the access point device **108-1** is a root node that configures the access point devices **108-2** and **108-3**. For example, a control device **112** provides configuration information to the access point device **108-1**, which uses the configuration information to configure itself and the access point devices **108-2** and **108-3**. The configuration information includes frequency spectrum information indicating a plurality of frequency bands to be used for different types of communications.

(20) In one or more implementations, the access point devices **108-1**, **108-2** and **108-3** are peers and are configured independently. For example, the control device **112** provides the configuration information to the access point devices **108-1**, **108-2**, and **108-3**, which use the configuration information to configure themselves.

(21) The control device **112** enables management of cellular services (e.g., 4G LTE and 5G NR services) and WLAN services (e.g., Wi-Fi services) in the private network **102** formed by the access point devices **108-1**, **108-2**, and **108-3**, which provide a 4G or 5G small cell. More particularly, the control device **112** enables management of devices, network configuration, Quality of Service (QoS) designations, and subscription services. For example, the control device **112** enables configuration of private cellular network parameters, including Public Land Mobile Network (PLMN) identifiers, configuration of access point devices to form a private cellular small cell, registration and provisioning of end user or client devices (e.g., User Equipment (UE) devices), management of Subscriber Identity Modules (SIMs) and International Mobile Subscriber

Identity numbers (IMSI), communication with a spectrum management system for private 4G and 5G networks, QoS designations for devices and applications, QoS measurement and reporting configuration, policy management and control, transport Protocol designations (e.g., WLAN only (e.g., Wi-Fi only), cellular only (e.g., 4G or 5G only), both WLAN and cellular) for devices and applications, management of subscription services, network connection and authorization management, management of phone numbers, an Application Programming Interface (API) to a public cellular network (e.g., operated by DISH Network), and associated user interface for a user/operator portal for the above features.

(22) In one or more implementations, the control device **112** enables an operator of the private network **102** to manage user profile settings and accounts of authorized users, and manage related subscriptions. Also, the control device **112** provides a dashboard of network health and Key Performance Indicators (KPIs) (e.g., peak data rate, peak spectral efficiency, data rate experienced by user, area traffic capacity, latency (User Plane), connection density, average spectral efficiency, energy efficiency, reliability, mobility, mobility interruption time, and bandwidth, which have been established and associated with specific minimum KPI values for different categories of services). Additionally, the control device **112** enables access point device and client device registration and policy settings, management/selection of licensed spectrum, management of transport protocol settings, management of phone numbers, management of QoS settings for devices and applications, and access point device configuration. In addition, the control device **112** interfaces with a 5G User Plane Function (UPF) of the 5G private network core **104** and also interfaces with a 5G Policy Control Function (PCF) of the 5G private network core **104**. Additionally or alternatively, the control device **112** can interface with a 4G UPF and 4G PCF.

(23) The communication system **100** also includes a channel allocation device **114** that operates in the private network core **104**. In one or more implementations, the communication system **100** includes an enterprise Local Area Network (LAN) **116**. In one or more implementations, the communication system **100** includes a network **118**, which includes one or more devices that provide voice and messaging services.

(24) The channel allocation device **114** performs Radio Unit (RU) channel aggregation orchestration. For example, the channel allocation device **114** manages channel aggregation using n48 and n77 frequency bands, by selecting and deploying a channel formed by aggregating n48(2A) and n77A frequency bands, or a channel formed by aggregating n48B and n77A frequency bands, across multiple access point devices. Also, the channel allocation device **114** performs real-time analysis of RU spectrum band use, performs real-time analysis of RU channel aggregation channel use (e.g., in n48 and n77 frequency bands), and performs real-time analysis of network payload, use case requirements, and/or QoS requirements. Additionally, the channel allocation device **114** interfaces with a spectrum management system for the private network **102**, and interfaces with a CBRS Spectrum Access Server (SAS) (not shown) that grants authorization to n48 licensed spectrum (PALs) to RUs. In addition, the channel allocation device **114** performs or interfaces with devices that perform RU output power orchestration.

(25) FIG. 2 is a block diagram illustrating an example of an access point device **108** in accordance with embodiments described herein. The access point devices **108-1**, **108-2**, and **108-3** shown in FIG. 1 have the same configuration as the access point device **108**. In some embodiments, one or more special-purpose computing systems may be used to implement the access point device **108**. Accordingly, various embodiments described herein may be implemented in software, hardware, firmware, or in some combination thereof. The access point device **108** may include one or more memory devices **204**, one or more central processing units (CPUs) **210**, I/O interfaces **212**, other computer-readable media **214**, and network interfaces **216**.

(26) The one or more memory devices **204** may include one or more various types of non-volatile and/or volatile storage technologies. Examples of the one or more memory devices **204** may include, but are not limited to, flash memory, hard disk drives, optical drives, solid-state drives,

various types of random access memory (RAM), various types of read-only memory (ROM), other computer-readable storage media (also referred to as processor-readable storage media), or the like, or any combination thereof. The one or more memory devices **204** may be utilized to store information, including computer-readable instructions that are utilized by the one or more CPUs **210** to perform actions, including those of embodiments described herein.

(27) The one or more memory devices **204** may have stored thereon an access point module **206**. The access point module **206** is configured to implement and/or perform some or all of the functions of the access point device **108** described herein. The one or more memory devices **204** may also store other programs and data **208**, which may include digital certificates, connection recovery algorithms, connection recovery rules, network protocols, O-RAN operating rules, user interfaces, operating systems, etc.

(28) I/O interfaces **212** may include enhanced Common Public Radio Interface (eCPRI) ports, Antenna Interface Standards Group (AISG) interfaces, other data input or output interfaces, or the like. Other computer-readable media **214** may include other types of stationary or removable computer-readable media, such as removable flash drives, external hard drives, or the like. Network interfaces **216** are configured to communicate with other computing devices including client devices and other access point devices. In various embodiments, the network interfaces **216** include transmitters and receivers, a layer 2 (L2) switch and physical network ports (not illustrated) to send and receive data as described herein, and to send and receive instructions, commands and data to implement the processes described herein. For example, some of the transmitters and receivers of the network interfaces **216** are configured to transmit and receive information according to IEEE 802.11 communication standards (e.g., 802.11, 802.11a, 802.11b, 802.11g, 802.11n, 802.11ac, 802.11ax, 802.11be, etc.), and some of the transmitters and receivers of the network interfaces **216** are configured to transmit and receive information according to cellular communication standards (e.g., 4G LTE, 5G, etc.).

(29) FIG. 3 is a block diagram illustrating an example of a control device **112** in accordance with embodiments described herein. In some embodiments, one or more special-purpose computing systems may be used to implement the control device **112**. Accordingly, various embodiments described herein may be implemented in software, hardware, firmware, or in some combination thereof. The control device **112** may include one or more memory devices **304**, one or more central processing units (CPUs) **310**, I/O interfaces **312**, other computer-readable media **314**, and network interfaces **316**.

(30) The one or more memory devices **304** may include one or more various types of non-volatile and/or volatile storage technologies. Examples of the one or more memory devices **304** may include, but are not limited to, flash memory, hard disk drives, optical drives, solid-state drives, various types RAM, various types of ROM, other computer-readable storage media (also referred to as processor-readable storage media), or the like, or any combination thereof. The one or more memory devices **304** may be utilized to store information, including computer-readable instructions that are utilized by the one or more CPUs **310** to perform actions, including those of embodiments described herein.

(31) The one or more memory devices **304** may have stored thereon a control module **306**. The control module **306** is configured to implement and/or perform some or all of the functions of the control device **112** described herein. The one or more memory devices **304** may also store other programs and data **308**, which may include programs for communicating with the channel allocation device **114** and the access point devices **108-1**, **108-2**, and **108-3**, digital certificates, network protocols, user interfaces, operating systems, etc.

(32) I/O interfaces **312** may include enhanced data input or output interfaces, or the like. Other computer-readable media **314** may include other types of stationary or removable computer-readable media, such as removable flash drives, external hard drives, or the like. Network interfaces **316** are configured to communicate with other computing devices including the channel

allocation device **114** and the access point devices **108-1**, **108-2**, and **108-3**. In various embodiments, the network interfaces **316** include transmitters and receivers, physical network ports (not illustrated) to send and receive data as described herein, and to send and receive instructions, commands and data to implement the processes described herein.

(33) FIG. **4** is a block diagram illustrating an example of a channel allocation device **114** in accordance with embodiments described herein. In some embodiments, one or more special-purpose computing systems may be used to implement the channel allocation device **114**.

Accordingly, various embodiments described herein may be implemented in software, hardware, firmware, or in some combination thereof. The channel allocation device **114** may include one or more memory devices **404**, one or more central processing units (CPUs) **410**, I/O interfaces **412**, other computer-readable media **414**, and network interfaces **416**.

(34) The one or more memory devices **404** may include one or more various types of non-volatile and/or volatile storage technologies. Examples of the one or more memory devices **404** may include, but are not limited to, flash memory, hard disk drives, optical drives, solid-state drives, various types RAM, various types of ROM, other computer-readable storage media (also referred to as processor-readable storage media), or the like, or any combination thereof. The one or more memory devices **404** may be utilized to store information, including computer-readable instructions that are utilized by the one or more CPUs **410** to perform actions, including those of embodiments described herein.

(35) The one or more memory devices **404** may have stored thereon a channel allocation module **406**. The channel allocation module **406** is configured to implement and/or perform some or all of the functions of the channel allocation device **114** described herein. The one or more memory devices **404** may also store other programs and data **408**, which may include digital certificates, network protocols, user interfaces, operating systems, etc., programs for communicating with the control device **112**, the access point devices **108-1**, **108-2**, and **108-3**, and a spectrum management system, which communicates with a Spectrum Access System (SAS) service provider device and a device that maintains databases of frequency bands for licenses that have already been acquired by an operator of the core network **104**.

(36) I/O interfaces **412** may include enhanced data input or output interfaces, or the like. Other computer-readable media **414** may include other types of stationary or removable computer-readable media, such as removable flash drives, external hard drives, or the like. Network interfaces **416** are configured to communicate with other computing devices including the control device **112**, the access point devices **108-1**, **108-2**, and **108-3**, Spectrum Access System (SAS) service provider devices, and devices that maintain databases of frequency bands for licenses that have already been acquired. In various embodiments, the network interfaces **416** include transmitters and receivers, physical network ports (not illustrated) to send and receive data as described herein, and to send and receive instructions, commands and data to implement the processes described herein.

(37) FIGS. **5A** and **5B** illustrate a logical flow diagram showing an example of a method **500** of operating a communication system in accordance with embodiments described herein. The method **500** begins at **502** in FIG. **5A**.

(38) At **502**, first channel requirement information indicating a first bandwidth of a channel is obtained. For example, at **502**, the channel allocation device **114** obtains the first channel requirement information indicating the first bandwidth of the channel from the control device **112**, which determines the first bandwidth of the channel based on configuration settings made by an operator of the private network **102** via operation of the control device **112**. The method **500** then proceeds to **504**.

(39) At **504**, available channel information indicating a plurality of first Citizens Broadband Radio Service (CBRS) frequency bands and a plurality of first non-CBRS frequency bands available to be allocated to the channel is obtained. For example, at **504**, the channel allocation device **114** obtains

the available channel information indicating the plurality of first CBRS frequency bands and the plurality of first non-CBRS frequency bands available to be allocated to the channel from a spectrum management device, which communicates with a spectrum management system for the private network **102** and a CBRS Spectrum Access Server (SAS). The method **500** then proceeds to **506**.

(40) At **506**, one or more of the first CBRS frequency bands is selected and one or more of the first non-CBRS frequency bands is selected. For example, at **506**, if the first bandwidth of the channel is 35 megahertz (MHz), the channel allocation device **114** selects a frequency band having a bandwidth of 20 MHz in the n48 frequency band and selects a frequency band having a bandwidth of 15 MHz in the n77 frequency band, so that the sum of the bandwidth of the selected frequency bands is the first bandwidth (e.g., 35 MHz) indicated by the first channel requirement information obtained at **502**. The method **500** then proceeds to **508**.

(41) At **508**, first channel aggregation information indicating the one or more of first CBRS frequency bands selected at **506** and the one or more of first non-CBRS frequency bands selected at **506** is generated. For example, at **508**, the channel allocation device **114** generates a message including an identifier of each of the selected frequency bands (e.g., in the n48(2A) and n77A frequency bands). The method **500** then proceeds to **510**.

(42) At **510**, first transmission power level information indicating one or more first transmission power levels respectively corresponding to the one or more of the first CBRS frequency bands and one or more first transmission power levels respectively corresponding to the one or more of the first non-CBRS frequency bands is generated. For example, at **510**, the channel allocation device **114** stores a table or other suitable data structure including identifiers of each of the frequency bands that is available for allocation to the channel in association with corresponding power level values that have been determined (e.g., calculated) previously. The method **500** then proceeds to **512**.

(43) At **512**, the first channel aggregation information generated at **508** and the first transmission power level information generated at **510** are transmitted to an external device. The method **500** then proceeds to **514**. For example, at **510**, the channel allocation device **114** transmits a message including the first channel aggregation information generated at **508** and the first transmission power level information generated at **510** to the control device **112**. By way of another example, at **510**, the channel allocation device **114** transmits a message including the first channel aggregation information generated at **508** and the first transmission power level information generated at **510** to the access point devices **108-1**, **108-2**, and **108-3**. The method **500** then proceeds to **514**.

(44) At **514**, the first channel aggregation information and the first transmission power level information transmitted at **512** is received by at least one access point device. For example, at **514**, the access point devices **108-1**, **108-2**, and **108-3** receive the first channel aggregation information and the first transmission power level information transmitted at **512**. The method **500** then proceeds to **516**.

(45) At **516**, first data is transmitted in the channel using the one or more of the first CBRS frequency bands selected at **506** and the one or more of the first non-CBRS frequency bands selected at **506** based on the first transmission power level information generated at **510**. For example, at **516**, the access point device **108-1** transmits the first data to the client device **110-1** in the channel using the one or more of the first CBRS frequency bands and the one or more of the first non-CBRS frequency bands based on the first transmission power level information received at **514**. The method **500** then proceeds to **518** in FIG. 5B.

(46) At **518**, second channel requirement information indicating a second bandwidth of a channel is obtained. For example, at **518**, the channel allocation device **114** obtains the second channel requirement information indicating the second bandwidth of the channel from the control device **112**, which determines the second bandwidth of the channel based on one or more parameters (e.g., throughput, bandwidth, or delay, etc.) calculated based on performance data obtained from within

the private network **102**, and a table or other data structures in which a plurality of identifiers of the one or more parameters is stored in association with a plurality of corresponding bandwidth values, which have been determined based on computer simulation. By way of another example, at **518**, the channel allocation device **114** obtains the second channel requirement information indicating the second bandwidth of the channel from the control device **112**, which determines the bandwidth of the channel based on configuration settings made by an operator of the private network **102**, for example, that is upgrading one or more service subscriptions so that the private network **102** can provide network access to a greater number of client devices. The method **500** then proceeds to **520**.

(47) At **520**, available channel information indicating a plurality of second CBRS frequency bands and a plurality of second non-CBRS frequency bands available to be allocated to the channel is obtained. For example, at **520**, the channel allocation device **114** obtains the available channel information indicating the plurality of second CBRS frequency bands and the plurality of second non-CBRS frequency bands available to be allocated to the channel from a spectrum management device, which communicates with a spectrum management system for the private network **102** and a CBRS Spectrum Access Server (SAS). The method **500** then proceeds to **522**.

(48) At **522**, one or more of the second CBRS frequency bands is selected and one or more of the second non-CBRS frequency bands is selected. For example, at **522**, if the second bandwidth of the channel is 45 megahertz (MHz), the channel allocation device **114** selects a frequency band having a bandwidth of 20 MHz in the n48 frequency band and selects a frequency band having a bandwidth of 25 MHz in the n77 frequency band, so that the sum of the bandwidth of the selected frequency bands is the second bandwidth (e.g., 45 MHz) indicated by the second channel requirement information obtained at **518**. The method **500** then proceeds to **524**.

(49) At **524**, second channel aggregation information indicating the one or more of second CBRS frequency bands selected at **522** and the one or more of second non-CBRS frequency bands selected at **522** is generated. For example, at **524**, the channel allocation device **114** generates a message including an identifier of each of the selected frequency bands (e.g., n48(2A) and n77A). The method **500** then proceeds to **526**.

(50) At **526**, second transmission power level information indicating one or more second transmission power levels respectively corresponding to the one or more of the second CBRS frequency bands and one or more second transmission power levels respectively corresponding to the one or more of the second non-CBRS frequency bands is generated. For example, at **526**, the channel allocation device **114** stores a table or other suitable data structure including identifiers of each of the frequency bands that is available for allocation to the channel in association with corresponding power level values that have been determined (e.g., calculated) previously. The method **500** then proceeds to **528**.

(51) At **528**, the second channel aggregation information generated at **524** and the second transmission power level information generated at **526** are transmitted to an external device. For example, at **528**, the channel allocation device **114** transmits a message including the second channel aggregation information generated at **524** and the second transmission power level information generated at **526** to the control device **112**. By way of another example, at **528**, the channel allocation device **114** transmits a message including the second channel aggregation information generated at **524** and the second transmission power level information generated at **526** to the access point devices **108-1**, **108-2**, and **108-3**. The method **500** then proceeds to **530**.

(52) At **530**, the second channel aggregation information and the second transmission power level information transmitted at **528** is received by at least one access point device. For example, at **532**, the access point devices **108-1**, **108-2**, and **108-3** receive the second channel aggregation information and the second transmission power level information transmitted at **528**. The method **500** then proceeds to **532**.

(53) At **532**, second data is transmitted in the channel using the one or more of the second CBRS

frequency bands selected at **522** and the one or more of the second non-CBRS frequency bands selected at **522** based on the second transmission power level information generated at **526**. For example, at **532**, the access point device **108-1** transmits the second data to the client device **110-1** in the channel using the one or more of the second CBRS frequency bands and the one or more of the second non-CBRS frequency bands based on the second transmission power level information received at **530**. The method **500** then ends.

(54) FIGS. **6A** and **6B** illustrate a logical flow diagram showing an example of a method **600** of operating a communication system in accordance with embodiments described herein. The method **600** begins at **602** in FIG. **6A**.

(55) At **602**, access point information indicating a plurality of access point devices is obtained, wherein the access point information indicates at least a first access point device and a second access point device. For example, at **602**, the control device **112** obtains the access point information in response to an operator of the control device **112** entering that information via a user interface provided by the control device **112**. The method **600** then proceeds to **604**.

(56) At **604**, client device information indicating a plurality of client devices is obtained, wherein the access point information indicates at least a first client device and a second client device. For example, at **604**, the control device **112** obtains the client device information in response to an operator of the control device **112** entering that information via a user interface provided by the control device **112**. The method **600** then proceeds to **606**.

(57) At **606**, client device requirement information indicating one or more requirements of the client devices is obtained. For example, at **606**, the control device **112** obtains the client device requirement indicating a minimum throughput or bandwidth value or a maximum latency value in response to an operator of the control device **112** entering those values via a user interface provided by the control device **112**. By way of another example, at **606**, the control device **112** generates the client device requirement indicating the minimum throughput or bandwidth value or the maximum latency value using a table or other suitable data structure storing a plurality values of requirements in association with a plurality of corresponding throughput, bandwidth, and/or latency values, or using a computer model and/or simulation, based on information entered via a user interface provided by the control device **112**. The method **600** then proceeds to **608**.

(58) At **608**, data transmission capability information indicating whether each of the client devices has a capability to transmit data according to a cellular communication standard (e.g., 4G, 5G), a Wireless Local Area Network (WLAN) communication standard (e.g., Wi-Fi-4, Wi-Fi-5, Wi-Fi-6, Wi-Fi-6E), or both a cellular communication standard and a WLAN communication standard is obtained. For example, at **608**, the control device **112** obtains the data transmission capability in response to an operator of the control device **112** entering that information via a user interface provided by the control device **112**. The method **600** then proceeds to **610**.

(59) At **610**, frequency band information indicating at least one frequency band usable by each of the client devices is obtained. For example, at **610**, the control device **112** stores a table or other suitable data structure including identifiers of each of a plurality of data transmission capability identifiers stored in association with corresponding identifiers of frequency bands, and obtains the frequency band information therefrom using the data transmission capability information obtained at **608**. By way of another example, at **610**, the client device obtains the frequency band information in response to an operator of the control device **112** entering those values via a user interface provided by the control device **112**. In one or more implementations, the control device **112** selects the at least one frequency band usable by a client device based on a PLMN identifier stored in a SIM profile for the client device, and a table or other suitable data structure in which a plurality of PLMN identifiers is stored in association with a plurality of corresponding identifiers of frequency bands that may be allocated and/or not allocated. The method **600** then proceeds to **612**.

(60) At **612**, application information indicating one or more applications is obtained. For example,

at **612**, the control device **112** obtains the application information (e.g., indicating a voice call application, a streaming video application, a security camera application, a VR/AR application, etc.) in response to an operator of the control device **112** entering that information via a user interface provided by the control device **112**. The method **600** then proceeds to **614**.

(61) At **614**, application requirement information indicating at least one requirement of the one or more applications is obtained. For example, at **614**, the control device **112** stores a table or other suitable data structure including identifiers of each of a plurality of application identifiers stored in association with corresponding requirement values, and obtains the application requirement information therefrom using the data application information obtained at **612**. By way of another example, at **614**, the control device **112** obtains the application requirement information in response to an operator of the control device **112** entering those values via a user interface provided by the control device **112**. The method **600** then proceeds to **616**.

(62) At **616**, channel information indicating a plurality of channels is obtained, wherein the channel information indicates at least a first channel, a second channel, and a third channel, along with corresponding bandwidth values for each channel. For example, at **616**, the control device **112** obtains the channel information in response to an operator of the control device **112** entering those values via a user interface provided by the control device **112**. By way of another example, at **616**, the control device **112** generates the channel information using a table or other suitable data structure storing channel identifiers in association with corresponding bandwidth values or using a computer model and/or simulation, based on information entered via a user interface provided by the control device **112**. The method **600** then proceeds to **618**.

(63) At **618**, the channel information for the first channel indicated by the channel information generated at **616** is transmitted to a channel allocation device. For example, at **618**, the control device **112** transmits the channel information for the first channel indicated by the channel information generated at **616** to the channel allocation device **114**. The method **600** then proceeds to **620** in FIG. 6B.

(64) At **620**, frequency band information indicating one or more frequency bands associated with the first channel is received from the channel allocation device. For example, at **620**, the control device **112** receives the frequency band information indicating one or more frequency bands associated with the first channel from the channel allocation device **114**. The method **600** then proceeds to **622**.

(65) At **622**, the channel information for the second channel indicated by the channel information generated at **616** is transmitted to the channel allocation device. For example, at **622**, the control device **112** transmits the channel information for the second channel indicated by the channel information generated at **616** to the channel allocation device **114**. The method **600** then proceeds to **624**.

(66) At **624**, frequency band information indicating one or more frequency bands associated with the second channel is received from the channel allocation device. For example, at **624**, the control device **112** receives the frequency band information indicating one or more frequency bands associated with the second channel from the channel allocation device **114**. The method **600** then proceeds to **622**.

(67) At **626**, the channel information for the third channel indicated by the channel information generated at **616** is transmitted to the channel allocation device. For example, at **626**, the control device **112** transmits the channel information for the third channel indicated by the channel information generated at **616** to the channel allocation device **114**. The method **600** then proceeds to **628**.

(68) At **628**, frequency band information indicating one or more frequency bands associated with the third channel is received from the channel allocation device. For example, at **628**, the control device **112** receives the frequency band information indicating one or more frequency bands associated with the third channel from the channel allocation device **114**. The method **600** then

proceeds to **630**.

(69) At **630**, first data is transmitted from a first access point device to a first client device according to a cellular communication standard using the one or more frequency bands associated with the first channel. For example, at **630**, the access point device **108-1** transmits the first data to the client device **110-1** according to a 3GPP 4G or 5G cellular communication standard using the one or more frequency bands associated with the first channel. The method **600** then proceeds to **632**.

(70) At **632**, second data is transmitted from the first access point device to a second client device according to a WLAN communication standard using the one or more frequency bands associated with the second channel. For example, at **632**, the access point device **108-1** transmits the second data to the client device **110-2** according to a Wi-Fi-6 communication standard using the one or more frequency bands associated with the second channel. The method **600** then proceeds to **634**.

(71) At **634**, third data is transmitted from the first access point device to a second access point device according to a cellular communication standard or a WLAN communication standard using the one or more frequency bands associated with the third channel. For example, at **634**, the access point device **108-1** transmits the third data to the access point device **108-2** according to a 3GPP 4G or 5G cellular standard using the one or more frequency bands associated with the third channel. The method **600** then ends.

(72) The various embodiments described above can be combined to provide further embodiments. Aspects of the embodiments can be modified, if necessary to employ concepts of the various patents, applications and publications to provide yet further embodiments.

(73) These and other changes can be made to the embodiments in light of the above-detailed description. In general, in the following claims, the terms used should not be construed to limit the claims to the specific embodiments disclosed in the specification and the claims, but should be construed to include all possible embodiments along with the full scope of equivalents to which such claims are entitled. Accordingly, the claims are not limited by the disclosure.

Claims

1. A method of operating a private network, the method including: obtaining, by a channel allocation device, first channel requirement information indicating a first bandwidth of a channel; obtaining, by the channel allocation device, available channel information indicating a plurality of first Citizens Broadband Radio Service (CBRS) frequency bands and a plurality of first non-CBRS frequency bands available to be allocated to the channel; selecting, by the channel allocation device, one or more of the first CBRS frequency bands and one or more of the first non-CBRS frequency bands based on the first channel requirement information; generating, by the channel allocation device, first channel aggregation information indicating the one or more of the first CBRS frequency bands and the one or more of the first non-CBRS frequency bands selected based on the first channel requirement information; generating, by the channel allocation device, first transmission power level information indicating one or more first transmission power levels respectively corresponding to the one or more of the first CBRS frequency bands and one or more first transmission power levels respectively corresponding to the one or more of the first non-CBRS frequency bands; transmitting, by the channel allocation device, to an external device, the first channel aggregation information and the first transmission power level information; receiving, by a first access point device, the first channel aggregation information and the first transmission power level information; and transmitting, by the first access point device, first data in the channel using the one or more of the first CBRS frequency bands and the one or more of the first non-CBRS frequency bands based on the first transmission power level information.

2. The method according to claim 1, wherein the one or more of the first CBRS frequency bands include one or more frequency bands in a range from 3550 to 3700 megahertz, and the one or more

of the first non-CBRS frequency bands include one or more frequency bands in a range from 3300 to 4200 megahertz.

3. The method according to claim 1, wherein the one or more of the first CBRS frequency bands include one or more frequency bands in a range from 3550 to 3700 megahertz, and the one or more of the first non-CBRS frequency bands include one or more frequency bands in a range from 2401 to 2495 megahertz, one or more frequency bands in a range from 5030 to 5990 megahertz, or one or more frequency bands in a range from 5925 to 7125 megahertz.

4. The method according to claim 3, wherein at least one of the one or more first transmission power levels respectively corresponding to the one or more of the first non-CBRS frequency bands is less than each of the one or more first transmission power levels respectively corresponding to the one or more of the first CBRS frequency bands.

5. The method according to claim 1, wherein the transmitting the first data includes transmitting the first data to a second access point device.

6. The method according to claim 1, wherein the transmitting the first data includes transmitting the first data to a client device.

7. The method according to claim 1, further comprising: obtaining, by the channel allocation device, second channel requirement information indicating a second bandwidth of the channel; obtaining, by the channel allocation device, available channel information indicating a plurality of second CBRS frequency bands and a plurality of second non-CBRS frequency bands available to be allocated to the channel; selecting, by the channel allocation device, one or more of the second CBRS frequency bands and one or more of the second non-CBRS frequency bands based on the second channel requirement information; generating, by the channel allocation device, second channel aggregation information indicating the one or more of the second CBRS frequency bands and the one or more of the second non-CBRS frequency bands selected based on the second channel requirement information; generating, by the channel allocation device, second transmission power level information indicating one or more second transmission power levels respectively corresponding to the one or more of the second CBRS frequency bands and one or more second transmission power levels respectively corresponding to the one or more of the second non-CBRS frequency bands; transmitting, by the channel allocation device, to the external device, the second channel aggregation information and the second transmission power level information; receiving, by the first access point device, the second channel aggregation information and the second transmission power level information; and transmitting, by the first access point device, second data in the channel using the one or more of the second CBRS frequency bands and the one or more of the second non-CBRS frequency bands based on the second transmission power level information.

8. The method according to claim 7, wherein the one or more of the first CBRS frequency bands are same as the one or more of the second CBRS frequency bands, the one or more of the first non-CBRS frequency bands are different from the one or more of the second non-CBRS frequency bands.

9. A system, comprising: a channel allocation device configured to: obtain first channel requirement information indicating a first bandwidth of a channel; obtain available channel information indicating a plurality of first Citizens Broadband Radio Service (CBRS) frequency bands and a plurality of first non-CBRS frequency bands available to be allocated to the channel; select one or more of the first CBRS frequency bands and one or more of the first non-CBRS frequency bands based on the first channel requirement information; generate first channel aggregation information indicating the one or more of the first CBRS frequency bands and the one or more of the first non-CBRS frequency bands selected based on the first channel requirement information; generate first transmission power level information indicating one or more first transmission power levels respectively corresponding to the one or more of the first CBRS frequency bands and one or more first transmission power levels respectively corresponding to the

one or more of the first non-CBRS frequency bands; and transmit, to an external device, the first channel aggregation information and the first transmission power level information; and a first access point device configured to: receive the first channel aggregation information and the first transmission power level information; and transmit first data in the channel using the one or more of the first CBRS frequency bands and the one or more of the first non-CBRS frequency bands based on the first transmission power level information.

10. The system according to claim 9, wherein the one or more of the first CBRS frequency bands include one or more frequency bands in a range from 3550 to 3700 megahertz, and the one or more of the first non-CBRS frequency bands include one or more frequency bands in a range from 3300 to 4200 megahertz.

11. The system according to claim 9, wherein the one or more of the first CBRS frequency bands include one or more frequency bands in a range from 3550 to 3700 megahertz, and the one or more of the first non-CBRS frequency bands include one or more frequency bands in a range from 2401 to 2495 megahertz, one or more frequency bands in a range from 5030 to 5990 megahertz, or one or more frequency bands in a range from 5925 to 7125 megahertz.

12. The system according to claim 9, wherein the first access point device transmits the first data by being configured to: transmit the first data to a second access point device.

13. The system according to claim 9, wherein the first access point device transmits the first data by being configured to: transmit the first data to a client device.

14. The system according to claim 9, wherein the channel allocation device is further configured to: obtain second channel requirement information indicating a second bandwidth of the channel; obtain available channel information indicating a plurality of second CBRS frequency bands and a plurality of second non-CBRS frequency bands available to be allocated to the channel; select one or more of the second CBRS frequency bands and one or more of the second non-CBRS frequency bands based on the second channel requirement information; generate second channel aggregation information indicating the one or more of the second CBRS frequency bands and the one or more of the second non-CBRS frequency bands selected based on the second channel requirement information; generate second transmission power level information indicating one or more second transmission power levels respectively corresponding to the one or more of the second CBRS frequency bands and one or more second transmission power levels respectively corresponding to the one or more of the second non-CBRS frequency bands; and transmit, to the external device, the second channel aggregation information and the second transmission power level information; and wherein the first access point device is further configured to: receive the second channel aggregation information and the second transmission power level information; and transmit second data in the channel using the one or more of the second CBRS frequency bands and the one or more of the second non-CBRS frequency bands based on the second transmission power level information.

15. The system according to claim 14, wherein the one or more of the first CBRS frequency bands are same as the one or more of the second CBRS frequency bands, the one or more of the first non-CBRS frequency bands are different from the one or more of the second non-CBRS frequency bands.

16. A non-transitory computer-readable storage medium having computer executable instructions stored thereon that, when executed by at least one processor, cause the at least one processor to perform actions, the actions comprising: obtaining first channel requirement information indicating a first bandwidth of a channel; obtaining available channel information indicating a plurality of first Citizens Broadband Radio Service (CBRS) frequency bands and a plurality of first non-CBRS frequency bands available to be allocated to the channel; selecting one or more of the first CBRS frequency bands and one or more of the first non-CBRS frequency bands based on the first channel requirement information; generating first channel aggregation information indicating the one or more of the first CBRS frequency bands and the one or more of the first non-CBRS frequency

bands selected based on the first channel requirement information; generating first transmission power level information indicating one or more first transmission power levels respectively corresponding to the one or more of the first CBRS frequency bands and one or more first transmission power levels respectively corresponding to the one or more of the first non-CBRS frequency bands; and transmitting to an external device, the first channel aggregation information and the first transmission power level information.

17. The non-transitory computer-readable storage medium according to claim 16, wherein the one or more of the first CBRS frequency bands include one or more frequency bands in a range from 3550 to 3700 megahertz, and the one or more of the first non-CBRS frequency bands include one or more frequency bands in a range from 3300 to 4200 megahertz.

18. The non-transitory computer-readable storage medium according to claim 16, wherein the one or more of the first CBRS frequency bands include one or more frequency bands in a range from 3550 to 3700 megahertz, and the one or more of the first non-CBRS frequency bands include one or more frequency bands in a range from 2401 to 2495 megahertz, one or more frequency bands in a range from 5030 to 5990 megahertz, or one or more frequency bands in a range from 5925 to 7125 megahertz.

19. The non-transitory computer-readable storage medium according to claim 18, wherein at least one of the one or more first transmission power levels respectively corresponding to the one or more of the first non-CBRS frequency bands is less than each of the one or more first transmission power levels respectively corresponding to the one or more of the first CBRS frequency bands.

20. The non-transitory computer-readable storage medium according to claim 16, wherein the computer executable instructions, when executed by the at least one processor, cause the at least one processor to perform further actions, the further actions comprising: obtaining second channel requirement information indicating a second bandwidth of the channel; obtaining available channel information indicating a plurality of second CBRS frequency bands and a plurality of second non-CBRS frequency bands available to be allocated to the channel; selecting one or more of the second CBRS frequency bands and one or more of the second non-CBRS frequency bands based on the second channel requirement information; generating second channel aggregation information indicating the one or more of the second CBRS frequency bands and the one or more of the second non-CBRS frequency bands selected based on the second channel requirement information; generating second transmission power level information indicating one or more second transmission power levels respectively corresponding to the one or more of the second CBRS frequency bands and one or more second transmission power levels respectively corresponding to the one or more of the second non-CBRS frequency bands; and transmitting, to the external device, the second channel aggregation information and the second transmission power level information.
